Program: **Diploma in Mechatronics**	
Course Code: **5439**	Course Title: **Automation and Control Lab**
Semester: **5**	Credits: **2**
Course Category: **Program Core**	
Periods per week: **3 (L:0, T:1, P:2)**	Periods per semester: **45**

Course Objectives:

- To provide hands-on training to construct electric power controllers using power semiconductor switching devices.
- To develop PLC ladder programs to realize real time applications.

Course Prerequisites:

Topic	Course code	Course name	Semester
Rectifiers	2039	Fundamentals of Electrical & Electronics Engineering Lab	2
Logic circuits	3048	Digital Electronics Lab	3

Course Outcomes:

On completion of the course, the students will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Design various gate triggering methods for SCR	9	Applying
CO2	Construct controlled rectifiers and power controlling circuits using power semiconductor switching devices	9	Applying
CO3	Implement power converter circuits using Power Semiconductor Switching Devices	12	Applying
CO4	Develop PLC ladder program for an application	6	Applying
	Lab Exam	6	
	Open Ended Project	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3		3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Construct the gate triggering methods for SCR**		
M1.01	Construct single phase control using resistancetriggering circuit and 1) find the maximum firing angle	3	Applying
M1.02	Construct single phase control using RCtriggering circuit and 1) find the maximum firing angle	3	Applying
M1.03	Construct single phase control using UJTtriggering circuit and 1) find the maximum firing angle	3	Applying
CO2	**Construct controlled rectifier circuits using power semiconductor switching devices**		
M2.01	Construct a single-phase half wave controlled rectifier using SCR with resistive load and 1) find firing angle	3	Applying
M2.02	Construct a single-phase full wave controlled rectifier using SCR with resistive load and 1) find firing angle	3	Applying
M2.03	Construct a single-phase bridge controlled rectifier using SCR with resistive load and 1) find firing angle	3	Applying
	Lab Exam 1	3	

CO3	**Implement industrial controlling circuits using power semiconductor switching devices**		
M3.01	Construct an illumination control circuit using DIAC and TRIAC	3	Applying
M 3.02	Implement an emergency lamp using SCR	3	Applying
M 3.03	Implement a batterycharger circuit using SCR	3	Applying
M3.04	Implement an inverter circuit using BJT and plot the waveform.	3	Applying
CO4	**Develop PLC ladder program for an application**		
M4.01	Develop ladder program for logic gates	1.5	Applying
M4.02	Develop ladder program for conveyor control	1.5	Applying
M4.03	Develop ladder program for water level control	1.5	Applying
M4.04	Develop ladder program for sevo motor speed control	1.5	Applying
	Open Ended Project	3	Applying
	Lab Exam 2	3	

Open Ended Projects

Not for End Semester Examination, but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 3-5. There should be no duplication in experiments between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on open ended experiment of their choice.

Example:

Automated Car Parking System
Automatic Vehicle Collision Prevention
System
Security System

Text /Reference:

T/R	Book Title/Author
T1	Industrial Electronics and Control - S K Bhattacharya, S Chatterjee.
T2	Programmable logic controllers - Frank D Petruzella

R1	Introduction to Programmable Logic Controllers - Gary Dunning - 3rd Edition –Delmar
R2	Power Electronics – P S Bimbhra

Online Resources:

Sl.No	Website Link
1	http://fac.ksu.edu.sa/sites/default/files/lab-manual_v3.pdf
2	http://onlinemas.weebly.com/uploads/6/3/9/4/6394050/lab_manual_shafeeq.pdf